

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B. Pharmacy

Semester: VIII

Subject Name: Biostatistics & Research Methodology

Subject Code: BP801T

Max Marks: 75

Date: 19/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)

(10 X 2) = 20

- i) What is the difference between Statistics and Biostatistics?
- ii) Describe Dispersion and Range
- iii) Enlist various online statistical software used in clinical trials.
- iv) Differentiate between null and alternative hypothesis.
- v) Define a) Normal distribution b) Poisson's distribution.
- vi) Enumerate different graphs used for representing quantitative data.
- vii) What is plagiarism?
- viii) Define factorial design and write their two advantages.
- ix) What is the difference between null hypothesis and alternative hypothesis?
- x) Explain multiple regression and standard error of regression.

Q.2. Long Answers (Answer 2 out of 3)

(10 X 2) = 20

- i) Describe different phases of clinical trials.
- ii) What is Curve Fitting? Explain curve fitting by the method of least squares.
- iii) From the following data calculate mean, median and mode.

Class limits	0-30	30-60	60-90	90-120	120-150	150-180
Frequency	8	13	22	27	18	7

Q.3. Short Answers (Answer 7 out of 9)

(5 X 7) = 35

- i) Enlist various non-parametric test and explain in detail Wilcoxon Rank Sum Test.
- ii) Explain t-test (Sample, Pooled or Unpaired and Paired).
- iii) Write a short note on historical design.
- iv) Discuss the applications of EXCEL and SPSS programmes in statistical analysis.
- v) Write a short note on a) Cohorts studies b) Observational studies.
- vi) Define hypothesis. Explain different types of hypothesis.
- vii) Write a note on Least Significance Difference.
- viii) What is sampling, write its objectives and explain the importance of sampling in hypothesis testing.
- ix) Describe experiential design technique.

*** END OF THE PAPER ***